

PRECISION IN ACTION
A cutting-edge lab at CG Semi's Sanand facility

cent of the cost. Companies will also remain free to export their products without restriction. "We took a leap of faith," says Sinha, "because chip-making is a strategic and foundational industry, with a whole range of sectors depending on reliable supply apart from quality products."

THE DOUBTING THOMASES

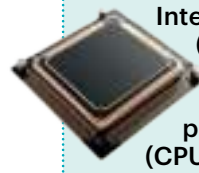
Not everyone, though, is convinced about the government's chip mission. Former Reserve Bank of India governor Raghuram Rajan called it a "ruinous race" for India to get into at this point, because besides being capital-intensive and heavy on subsidies, it is lean on labour employment. Instead of high-end chip manufacturing, Rajan thinks grants would be better spent on training the workforce in chip design, thereby strengthening India's R&D and talent pool. Persons of Indian origin already make up 20 per cent of the global workforce designing chips for multinational giants.

Kathir Thandavarayan, partner, Deloitte South Asia, begs to differ. "The government is making the right move," he says. "It is imperative that India, which is growing very fast and is likely to be a $10 trillion economy by 2035, has a semiconductor ecosystem. Any external dependency will always be disadvantageous. We need to build resilience and self-reliance by setting up semiconductor manufac-

KNOW YOUR CHIPS

These unseen workhorses power our digital lives, from the smartphone in your hand to the complex systems behind the Internet of Everything

MICROPROCESSORS



Integrated circuits (ICs) that contain one or more central processing units (CPUs) to execute instructions, serving as the core 'brains' of computing devices

GPUs



Specialised processors for massive parallel computation, Graphics Processing Units are critical not just for rendering graphics but now for powering advanced AI models

MEMORY CHIPS



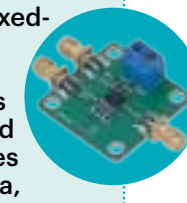
DRAM, NAND Flash, ROM and other memory types that store data temporarily or permanently, shaping the speed, capacity and performance of devices from smartphones to AI systems

ASICs



Application-Specific ICs are custom-built chips for specialised functions such as cryptocurrency mining, network processing or industrial automation

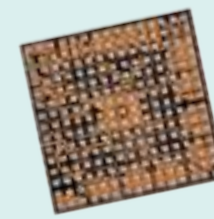
SIGNAL CONVERSION CHIP



Analog and mixed-signal ICs that transform physical inputs like light, sound and radio waves into digital data, vital for telecom, medical devices and autonomous vehicles

POWER MANAGEMENT CHIPS

Circuits that regulate and distribute power efficiently, ensuring safe, reliable operation across devices ranging from smartphones to EVs and cloud servers



SENSOR CHIPS



Components that detect physical phenomena—motion, temperature, pressure, light—powering automation, robotics, driver-assistance systems and wearable technology

